Exp 5 : Simulation and Throughput Analysis of LTE-Advanced Carrier Aggregation

Aim :

To simulate an LTE-Advanced wireless link using carrier aggregation and analyze the effect of aggregating multiple component carriers on available transmission bandwidth and estimated throughput using MATLAB.

Procedure

1. Open MATLAB and create a new script.
2. Define the component-carrier bandwidth, spectral efficiency and number of component carriers.
3. Calculate the single-carrier throughput using bandwidth × spectral efficiency.
4. Display the bandwidth and estimated throughput.
5. Plot the single-carrier result.

objective 1: Model One Component Carrier

Procedure :

1. Open MATLAB and create a new script.
2. Define the component-carrier bandwidth, spectral efficiency and number of component carriers.
3. Calculate the single-carrier throughput using bandwidth × spectral efficiency.
4. Display the bandwidth and estimated throughput.
5. Plot the single-carrier result.

Code :

```
clc;
clear;
close all;

%% 1. Define Parameters
BW1 = 20;       % Component carrier bandwidth (MHz)
SE = 4;         % Spectral efficiency (bits/s/Hz)
Ncc = 1;        % Number of component carriers

%% 2. Calculate Aggregate Bandwidth and Throughput
BWagg = BW1 * Ncc;
Throughput = BWagg * SE;  % Estimated throughput (Mbps)
```

```
%% 3. Display Results
disp('LTE-Advanced Single Carrier Throughput');
```

```
LTE-Advanced Single Carrier Throughput
```

```
disp('-------------------------------------');
```

```
-----------------------------------
```

```
disp(['Number of Component Carriers = ', num2str(Ncc)]);
```

```
Number of Component Carriers = 1
```

```
disp(['Bandwidth = ', num2str(BWagg), ' MHz']);
```

```
Bandwidth = 20 MHz
```

```
disp(['Spectral Efficiency = ', num2str(SE), ' bits/s/Hz']);
```

```
Spectral Efficiency = 4 bits/s/Hz
```

```
disp(['Estimated Throughput = ', num2str(Throughput), ' Mbps']);
```

```
Estimated Throughput = 80 Mbps
```

```
%% 4. Plot Estimated Throughput
figure('Color', 'w');

bar(Ncc, Throughput, 0.5);

grid on;
xlabel('Number of Component Carriers');
ylabel('Estimated Throughput (Mbps)');
title('LTE-Advanced Single Carrier Throughput');

% Set axis limits
xlim([0 2]);
ylim([0 100]);

% Display throughput value above the bar
text(Ncc, Throughput + 2, ...
    [num2str(Throughput), ' Mbps'], ...
    'HorizontalAlignment', 'center', ...
    'FontWeight', 'bold');

%% 5. Expected Output
fprintf('\nExpected Throughput: %.2f Mbps\n', Throughput);
```

```
Expected Throughput: 80.00 Mbps
```

Objective 2 : Extend to Multiple Aggregated Component Carriers

Procedure :

6. Define one, two and three component-carrier configurations.

7. Calculate total aggregated bandwidth for each configuration.

8. Estimate throughput using the same spectral-efficiency value.

9. Plot throughput for the different aggregation configurations.

10. Compare the results and explain the increase.

Code :

```
clc;
clear;
close all;

%% 1. Define Parameters
BWcc = 20;              % Bandwidth per component carrier (MHz)
SE = 4;                 % Spectral efficiency (bits/s/Hz)
CC = [1 2 3];           % Number of component carriers

%% 2. Calculate Aggregate Bandwidth and Throughput
AggBW = BWcc .* CC;             % Aggregate bandwidth (MHz)
Throughput = AggBW .* SE;       % Estimated throughput (Mbps)

%% 3. Display Results
disp('LTE-Advanced Carrier Aggregation Results');
```

```
LTE-Advanced Carrier Aggregation Results
```

```
disp('----------------------------------------');
```

```
----------------------------------------
```

```
disp('CC     Aggregate BW (MHz)     Throughput (Mbps)');
```

```
CC     Aggregate BW (MHz)     Throughput (Mbps)
```

```

disp([CC(:), AggBW(:), Throughput(:)]);
```

```
     1    20    80
     2    40   160
     3    60   240
```

```

%% 4. Plot Throughput
figure('Color', 'w');

bar(CC, Throughput, 0.6);

grid on;
xlabel('Number of Aggregated Component Carriers');
```

```
ylabel('Estimated Throughput (Mbps)');
title('LTE-Advanced Carrier Aggregation Throughput');

% Set axis limits and ticks
xticks(CC);
xlim([0 4]);
ylim([0 280]);

% Display throughput values above the bars
for k = 1:length(CC)
    text(CC(k), Throughput(k) + 5, ...
        sprintf('%d Mbps', Throughput(k)), ...
        'HorizontalAlignment', 'center', ...
        'FontWeight', 'bold');
end

%% 5. Display Final Results
fprintf('\nEstimated Throughput:\n');
```

```
Estimated Throughput:
```

```
for k = 1:length(CC)
    fprintf('%d carrier(s): %d MHz, %d Mbps\n', ...
        CC(k), AggBW(k), Throughput(k));
end
```

```
1 carrier(s): 20 MHz, 80 Mbps
2 carrier(s): 40 MHz, 160 Mbps
3 carrier(s): 60 MHz, 240 Mbps
```

Objective 3 : Compare Estimated Throughput for DIfferent Carrier Bandwidths Procedure

Procedure :

1. Define several component-carrier bandwidths.

2. Calculate throughput for each bandwidth using the selected spectral efficiency.

3. Compare the throughput values.

4. Plot bandwidth against estimated throughput.

5. Identify the bandwidth configuration that provides the highest estimated throughput.

MATLAB Program :

```
clc;
clear;
close all;

%% 1. Define Parameters
BW = [5 10 15 20];    % Carrier bandwidths (MHz)
SE = 4;               % Spectral efficiency (bits/s/Hz)
```

```
%% 2. Calculate Estimated Throughput
TP = BW .* SE;        % Estimated throughput (Mbps)

%% 3. Display Results
disp('Throughput vs Component Carrier Bandwidth');
```

```
Throughput vs Component Carrier Bandwidth
```

```
disp('----------------------------------------');
```

```
----------------------------------------
```

```
disp('Bandwidth (MHz)    Throughput (Mbps)');
```

```
Bandwidth (MHz)    Throughput (Mbps)
```

```
disp([BW(:), TP(:)]);
```

```
     5    20
    10    40
    15    60
    20    80
```

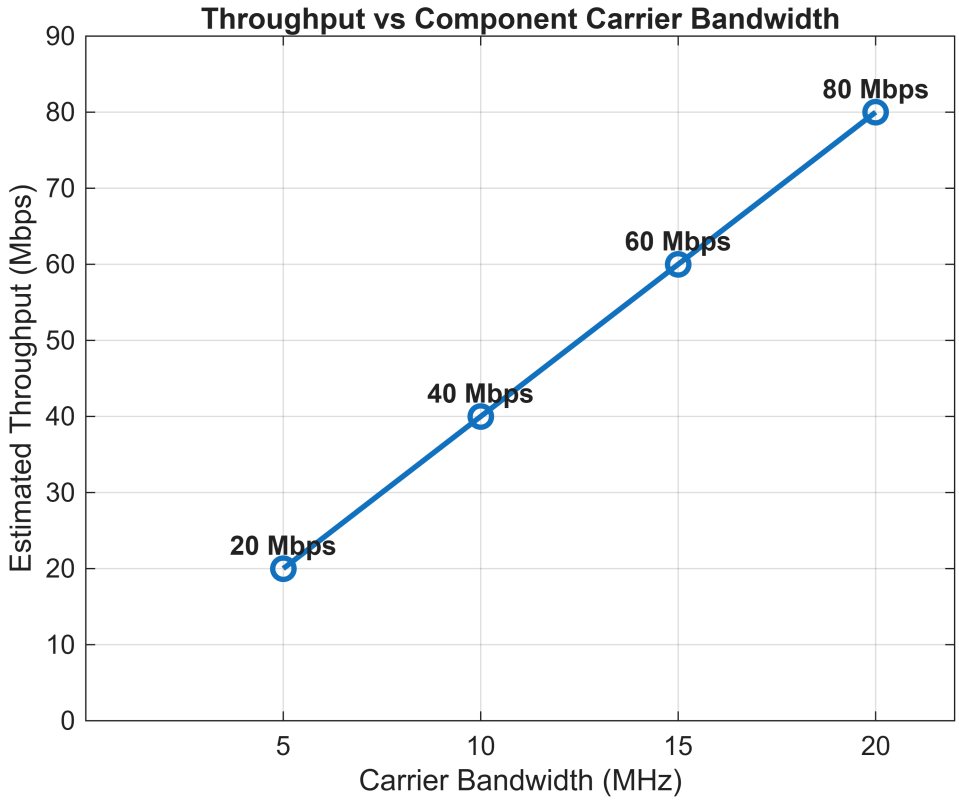
```
%% 4. Plot Throughput vs Bandwidth
figure('Color', 'w');

plot(BW, TP, 'o-', 'LineWidth', 2, 'MarkerSize', 8);

grid on;
xlabel('Carrier Bandwidth (MHz)');
ylabel('Estimated Throughput (Mbps)');
title('Throughput vs Component Carrier Bandwidth');

% Set axis ticks and limits
xticks(BW);
xlim([0 22]);
ylim([0 90]);

% Display throughput values next to the data points
for k = 1:length(BW)
    text(BW(k), TP(k) + 3, ...
        sprintf('%d Mbps', TP(k)), ...
        'HorizontalAlignment', 'center', ...
        'FontWeight', 'bold');
end
```




```
%% 5. Print Final Results
fprintf('\nEstimated Throughput Results:\n');
```

```
Estimated Throughput Results:
```

```
for k = 1:length(BW)
    fprintf('%d MHz bandwidth: %d Mbps\n', BW(k), TP(k));
end
```

```
5 MHz bandwidth: 20 Mbps
10 MHz bandwidth: 40 Mbps
15 MHz bandwidth: 60 Mbps
20 MHz bandwidth: 80 Mbps
```

Objective 4- Explain the Effect of Carrier Aggregation :

The total available bandwidth is obtained by combining the component carriers. Under the simplified constant spectral-efficiency model, throughput increases approximately in proportion to the aggregated bandwidth. In a practical LTE-Advanced system, the actual throughput is also influenced by modulation, coding, channel quality, control overhead, scheduling and radio conditions.

Result :

1. The LTE-Advanced link was successfully modeled using a single component carrier.

2. Multiple component carriers were aggregated and the resulting bandwidth and estimated throughput were calculated.

3. Throughput was compared for different carrier bandwidths.

4. The simulation demonstrated that carrier aggregation increases available transmission bandwidth and can increase achievable throughpu

```
%% Date and Time of Execution

currentTime = datetime('now', ...
    'Format', 'dd-MM-yyyy HH:mm:ss');

fprintf('\n=====================================\n');
```

```
=================================
```

```
fprintf('MATLAB Simulation Execution Details\n');
```

```
MATLAB Simulation Execution Details
```

```
fprintf('Date and Time: %s\n', char(currentTime));
```

```
Date and Time: 20-10-2026 09:46:57
```

```
fprintf('=====================================\n');
```

```
=================================
```